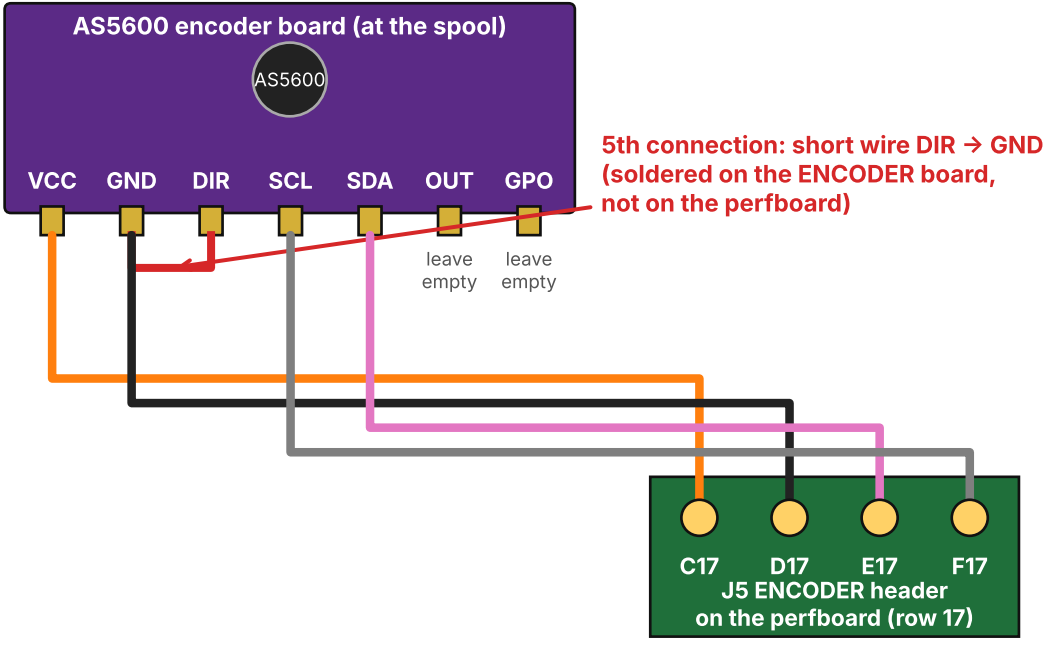
AS5600 encoder wiring (Rev D station): 4 wires to the board + DIR tied to GND on the encoder



Encoder pin	wire	perfboard J5	breadboard	signal
VCC	orange	C17	T+ (top red rail)	3.3 V
GND	black	D17	T- (top blue rail)	ground
SDA	pink	E17	a9	I2C data (XIAO D4 / ESP32 D21)
SCL	grey	F17	a10	I2C clock (XIAO D5 / ESP32 D22)
DIR	short	-	-	to the encoder's own GND pin (on the encoder)
OUT, GPO	-	-	-	not connected

Pin order differs between AS5600 boards: match the NAMES printed on your board, not the positions.
Why DIR: it sets the counting direction and must not float; a floating DIR can flip the sign at random.
Check: in the bench test, 'i' finds a device at 0x36 and 's' says magnet DETECTED.